SODIUM CHLORIDE 0.9%

Isotonic crystalloid

Hyperchloraemic acidosis (large volumes)

CHEMICAL NAME NaCl 9 g/L in water — Na^+ 154 mmol/L, Cl^- 154 mmol/L; osmolarity ~308 mOsm/L ("normal saline")

CLASS Isotonic crystalloid intravenous fluid

TRADE NAMES Normal saline · NS 0.9% · Sodium chloride 0.9% infusion

THERAPEUTIC USES / INDICATIONS

- Fluid resuscitation (hypovolaemia, shock, sepsis; balanced crystalloids often preferred).
- Replacement of losses (vomiting — with K^+; NG losses; hypochloraemic alkalosis).
- Maintenance (with added K^+ and glucose as appropriate).
- Diluent for IV drugs; line flushing; DKA initial resuscitation; hyponatraemia (mild hypovolaemic).

ADMINISTRATION

- **IV** via peripheral or central line; rate according to prescription and clinical status.
- Warm fluids in massive resuscitation/hypothermia.
- Label additives clearly (e.g. KCl).
- Bolus: 250–500 mL over <15 min with reassessment.

DOSAGES

- **Resuscitation bolus:** 500 mL over <15 min (250 mL in heart failure/elderly); children 10–20 mL/kg; reassess after each bolus.
- **Maintenance (adult):** 25–30 mL/kg/day total fluid (Na^+ 1 mmol/kg/day, K^+ 1 mmol/kg/day, glucose 50–100 g/day).
- **Children maintenance:** Holliday–Segar (100/50/20 mL/kg/day) — isotonic fluids recommended.

PHARMACOKINETICS

- Distributes throughout extracellular fluid (~25% remains intravascular after 1 h).
- Excess Na^+ and Cl^- excreted by kidneys.

PHARMACODYNAMICS

- Expands **extracellular and intravascular volume**; provides sodium and chloride.
- High chloride → ↓ strong ion difference → hyperchloraemic metabolic acidosis with large volumes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis at cannula site.
- Peripheral oedema with excess.

ADVERSE EFFECTS (SERIOUS)

- Fluid overload / pulmonary oedema (heart failure, renal failure, elderly).
- Hyperchloraemic metabolic acidosis; possible AKI with large volumes.
- Hypernatraemia (in water deficit).
- Dilutional coagulopathy (massive volumes).

CONTRAINDICATIONS & CAUTIONS

- Fluid overload; hypernatraemia.
- **Caution:** heart failure, oedematous states (cirrhosis, nephrotic syndrome), severe renal impairment, hyperchloraemia.

KEY INTERACTIONS

- **Lithium:** sodium load ↑ lithium excretion.
- Compatible as diluent with most drugs (not amphotericin, some others; check).

NURSING MANAGEMENT

- Prescription check: fluid, volume, rate, additives.
- Fluid balance chart, daily weight, vital signs, urine output (≥0.5 mL/kg/h).
- Assess for overload: breathlessness, crackles, raised JVP, oedema.
- Monitor U&E, chloride, bicarbonate daily on IV fluids.
- Cannula site care (VIP score).

PATIENT EDUCATION

- Report breathlessness, swelling, pain or redness at the drip site.

KEY POINTS

- Isotonic saline — Na^+/Cl^- 154 mmol/L.
- Hyperchloraemic acidosis with large volumes.
- Bolus 500 mL (<15 min) adults; 10–20 mL/kg children.
- Monitor fluid balance and U&E.

SOURCES CHECKED [953] Sodium Chloride 0.9 % w/v and Glucose 5 % w/v Intravenous Infusion BP (UK SmPC) [954] NICE CG174 [381] Evans et al., Intensive Care Med 2021 [588] Umpierrez et al., Diabetologia 2024 [955] NICE NG29 [587] JBDS-IP DKA guideline (2023)

Full references p. 432–457